

Albert Gassol Puigjaner

Email: albertgassol1@gmail.com

Mobile: +34 634 528 934

+47 929 39 948

Github: github.com/albertgassol1

Website: <https://albertgassol1.github.io/>

Linkedin: <https://www.linkedin.com/in/albert-gassol-puigjaner>

EDUCATION

- **Autonomous Robots Lab NTNU** Trondheim, Norway
PhD Candidate on Multi-modal Scene Understanding and Reasoning September 2024 - Ongoing
- **ETH Zurich** Zurich, Switzerland
MSc in Robotics, Systems and Control; GPA: 5.6/6 September 2021 - November 2023
Focus: Computer Vision and Machine Learning for Robotics
Master's Thesis: Learning Neural Vector Fields for Implicit Surfaces and Multi-view Reconstruction
- **Universitat Politècnica de Catalunya** Barcelona, Spain
Bachelor in Telecommunications Engineering; GPA: 8.98/10, Ranked first of class September 2017 - June 2021
Honours in 15 courses.
Bachelor's Thesis: High-performance Model Predictive Control and Graph-based Simultaneous Localization and Mapping for an autonomous Formula Student race-car

EXPERIENCE

- **Magic Leap** Zurich, Switzerland
Computer Vision Research Intern (Full-time) November 2023 - May 2024
 - **Augmented Reality without Borders: Achieving Precise Localization Without Maps:** Research on vision-based map-free localization leveraging open-source geo-referenced imagery data.
 - * Pre-print: <https://arxiv.org/abs/2408.17373>
- **Intelligent Control Systems Lab** Zurich, Switzerland
Research Assistant (Part-time) April 2023 - August 2023
 - **Coat-MPC: Performance-driven Constrained Optimal Auto-Tuner for MPC:** Research on goal-oriented safe Bayesian Optimization for MPC, with applications in autonomous racing.
 - * Repo: https://github.com/albertgassol1/coat_mpc
 - * Website: https://albertgassol1.github.io/coat_mpc
- **Motional** Singapore
Semantic Map Research Intern (Full-time) September 2022 - March 2023
 - **Traffic light detection:** Sensor fusion between camera and LiDAR accumulated point clouds for traffic light detection (deployed in production).
 - **LiDAR to camera targetless calibration:** LiDAR to camera targetless calibration using pole-like objects and semantic segmentation masks.

PUBLICATIONS

1. Albert Gassol Puigjaner, Manish Prajapat, Andrea Carron, Andreas Krause, and Melanie N. Zeilinger. Coat-mpc: Performance-driven constrained optimal auto-tuner for mpc. In *IEEE Robotics and Automation Letters (RA-L)*, 2025. <https://ieeexplore.ieee.org/document/10924398>
2. Albert Gassol Puigjaner, Angelos Zacharia, and Kostas Alexis. Relationship-aware hierarchical 3d scene graph. In *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026

FULL-TIME PROJECTS

- **Image Communication and Understanding, Computer Vision Lab at ETH** Zurich, Switzerland
Master's Thesis April 2023 - October 2023
 - **VF-NeRF: Learning Neural Vector Fields for Indoor Scene Reconstruction:** Novel multi-view neural surface reconstruction method that models the scene geometry as unit vectors directed towards the nearest surface. State-of-the-art results in 3D reconstruction of indoor scenes.
 - * Pre-print: <https://arxiv.org/abs/2408.08766>
 - * Repo <https://github.com/albertgassol1/vf-nerf>
- **AMZ Racing** Zurich, Switzerland
Planning and Control Engineer, Formula Student September 2021 - August 2022
 - **Performance-driven Bayesian Optimization for MPC tuning:** ETH semester project.
 - **MPC for unknown tracks:** Dynamic bicycle model MPC in spatial curvilinear coordinates with curvature minimization.

- **Lap time optimizer:** Lap time optimizer using a 4-wheel model.

- **BCN eMotorsport**

Barcelona, Spain

Autonomous Systems Lead / Planning and Control Engineer, Formula Student

September 2019 - August 2021

- **Lead of Perception, Estimation and Control:** Led a team of 10 engineers developing the autonomous system software stack of a driverless racing car from scratch. Season results:
https://www.youtube.com/watch?v=ly_ax8w-T7E&t=1s
- **Graph-based SLAM:** Graph SLAM for online localization and race track mapping.
- **MPC for unknown and known tracks:** Dynamic bicycle model MPC in cartesian and curvilinear coordinates.
- **Formula Student Simulator:** Tailored a race car simulator to test our autonomous stack.
- **Track boundary estimation and trajectory planning:** Geometrical approach using Voronoi Diagrams.

HONORS AND AWARDS

- **Ranked first of my Bachelor's class out of 300 students.**
- **Top average Bachelor's degree qualification.** Among the top 2% of all the students of my Bachelor's degree during the last 5 years.
- **Formula Student Germany 2022 Engineering Design Winner.**

SKILLS SUMMARY

- **Languages:** Catalan (Native), Spanish (Native), English (C1 and TOEFL 114/120), Norwegian (A1-A2)
- **Programming Languages:** C++, Python, C, Matlab, Bash, Java
- **Relevant Libraries:** ROS/ROS2, PyTorch, OpenCV, PyColmap, BoTorch, Open3D, PCL, Eigen, IpOpt, g2o, FastAPI, gRPC
- **Relevant Developer Tools:** Git, CMake, Bazel, Docker, AWS, ArgoWorkflow, GCP, NVIDIA Brev